Lösung: Höhere Ableitungen

$$f(x) = e^{-x} \cdot \sin(x)$$

$$\begin{aligned} f'(x) &= (e^{-x})' \cdot \sin(x) + e^{-x} \cdot (\sin(x))' \\ &= -e^{-x} \cdot \sin(x) + e^{-x} \cdot \cos(x) \\ &= e^{-x} (\cos(x) - \sin(x)) \end{aligned}$$

$$\begin{aligned} f''(x) &= (e^{-x})' \cdot (\cos(x) - \sin(x)) + e^{-x} (\cos(x) - \sin(x))' \\ &= -e^{-x} \cdot (\cos(x) - \sin(x)) + e^{-x} (-\sin(x) - \cos(x)) \\ &= e^{-x} (-\cos(x) + \sin(x) - \sin(x) - \cos(x)) \\ &= -2 e^{-x} \cdot \cos(x) \end{aligned}$$

$$\begin{aligned} f'''(x) &= 2 (-e^{-x})' \cdot \cos(x) + 2 (-e^{-x}) \cdot (\cos(x))' \\ &= 2 e^{-x} \cdot \cos(x) - 2 e^{-x} \cdot (-\sin(x)) \\ &= 2 e^{-x} (\cos(x) + \sin(x)) \end{aligned}$$